Earnings Through the Stages: Using Tax Data to Test for Sources of Error in CPS ASEC Earnings and Inequality Measures

by

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Abstract

In this paper, I explore the impact of generalized coverage error, item non-response bias, and measurement error on measures of earnings and earnings inequality in the CPS ASEC. I match addresses selected for the CPS ASEC to administrative data from 1040 tax returns. I then compare earnings statistics in the tax data for wage and salary earnings in samples corresponding to seven stages of the CPS ASEC survey production process. I also compare the statistics using the actual survey responses. The statistics I examine include mean earnings, the Gini coefficient, percentile earnings shares, and shares of the survey weight for a range of percentiles. I examine how the accuracy of the statistics calculated using the survey data is affected by including imputed responses for both those who did not respond to the full CPS ASEC and those who did not respond to the earnings question. I find that generalized coverage error and item nonresponse bias are dominated by measurement error, and that an important aspect of measurement error is households reporting no wage and salary earnings in the CPS ASEC when there are such earnings in the tax data. I find that the CPS ASEC sample misses earnings at the high end of the distribution from the initial selection stage and that the final survey weights exacerbate this.

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Introduction

This paper seeks to bridge the gap between our understanding of earnings and earnings inequality statistics estimated in administrative records and in survey data. In particular, I compare the Current Population Survey Annual Social and Economic Supplement (CPS ASEC) to tax data from the universe of 1040 tax filings. Over the last few years, a large and influential literature has used administrative records, mainly tax data, to explore different aspects of earnings and income in the US.[1] At the same time, many researchers, and government agencies rely on survey data, including data from the CPS ASEC, to create earnings statistics. This includes official statistics from the Congressional Budget Office and the Census Bureau (Guzman & Kollar, 2023; Shrider & Creamer, 2023; Habib, 2023).

Matching the CPS ASEC to tax data allows me to estimate earnings statistics at eight stages in the survey production process and compare them to the same statistics in the tax data. The differences in the statistics at each stage allow me to identify generalized coverage error, item nonresponse error, and measurement error in the CPS ASEC.

This work is made more important by prior research that has shown a decline in the quality of some earnings and inequality statistics based on household surveys including the CPS ASEC (Meyer, Mok, & Sullivan, 2015; Rothbaum, 2015; Bee, Gathright, & Meyer, 2015; Bollinger, Hirsch, Hokayem, & Ziliak, 2019). This paper adds to the growing effort to use administrative data to assess and better understand the strengths and weaknesses of survey data for earnings and inequality measurement.

There is a large and growing literature comparing administrative records to survey responses. Two such papers focus on how nonresponse bias affects earnings inequality in the CPS ASEC. A paper by Bee, Gathright, and Meyer matched the 2011 CPS ASEC to 1040 tax data and found little evidence of differences in the percentiles of the earnings distribution for survey respondents and nonrespondents (Bee, Gathright, & Meyer, 2015). More recently, Bollinger et al. matched data from the CPS ASEC to the Social Security Administration's Detailed Earnings Record (DER) to examine how item nonresponse affects estimates of earnings gaps (Bollinger, Hirsch, Hokayem, & Ziliak, 2019). They found that nonresponse is most common at the ends of the earnings distribution and that this causes the Gini coefficient to be biased downwards when calculated with data from the CPS ASEC. There are several key differences between Bollinger et al. and this paper. In this paper I look at not just nonresponse but several steps in the sampling process and measurement error. I also am able to use the universe of 1040s which allows me to make a comparison to the population of tax filers.

Several of the papers that use administrative data to assess the quality of survey data have examined the quality of imputations. The most relevant is the Bollinger et al. study mentioned above (Bollinger, Hirsch, Hokayem, & Ziliak, 2019). They found that including imputations helped to mitigate some of this bias caused by nonresponse at the tails of the distribution. Papers looking at other outcomes came to the opposite conclusion. Meyer et al. looked at imputed receipt of SNAP benefits and found that imputation led to more error in the CPS (Meyer, Mittag, & George, 2020). Hokayem et al. found that item nonresponse bias and imputation in the CPS ASEC leads to the CPS ASEC underestimating the poverty rate by about one percent (Hokayem, Bollinger, & Ziliak, 2015).

[1] A few of the many papers that this literature includes are (Saez & Zucman, 2020; Piketty, Saez, & Zucman, 2017; Auten & Splinter, 2019; Larrimore, Mortenson, & Splinter, 2020)

This paper's strategy of looking for differences in the estimates at different stages in the survey production processes is related to a paper by Meyer and Mittag. They decompose total survey error into generalized coverage error, item nonresponse error, and measurement error when looking at reported receipt of two large government transfer programs (Meyer & Mittag, 2021). They also further decompose item nonresponse error and measurement error to look at imputation errors and false positives and false negatives in reporting program receipt. They find that measurement error is much larger than generalized coverage error and item nonresponse. My paper differs from this approach in several ways. Most importantly, I look at earnings statistics rather than program receipt. Earnings statistics are important for us to study, but they have several disadvantages in this context. The statistics I use are not decomposable such that one can partition how much of them is due to different sub-populations and may not be as accurate in the administrative data. This is because taxes are subject to possible misreporting while Meyer and Mittag could use actual program payments. Another difference is that I separate generalized coverage error into seven stages to look more closely for where there could be error in the sampling process.[2]

To fix concepts, I define generalized coverage error, item nonresponse error, and measurement error. Generalized coverage error is bias coming from the survey frame not accurately reflecting the chosen population, weight adjustment, and unit nonresponse error. Item nonresponse error is bias caused by nonrandom households not responding the income questions on survey. Measurement error is error coming from how inaccurately respondents answer questions.

I focus on only wage and salary income (WSI) as CPS ASEC respondents are not directly asked for their adjusted gross income (AGI). Instead, AGI is recreated in the CPS ASEC based on a set of assumptions. For looking at measurement error, I want to compare what respondents say directly. The statistics I look at are mean earnings, the Gini coefficient, earnings shares, and the portion of the survey weight in different earnings percentiles.

I find throughout my analysis that there is generalized coverage error and item nonresponse bias, but they are swamped by measurement error. Addresses that have wage and salary earnings in the tax data but not in the CPS ASEC are an important part of that measurement error. I find that the CPS ASEC sample misses households at the top of the earnings distribution even from the initial selection stage. The final weights also appear to down-weight the top of the earnings distribution. Lastly, I find that imputed values bring some statistics closer to the tax filings and some farther.

1. Data

This paper combines data from two sources. The first is the CPS ASEC for survey years 2017 to 2019. The CPS ASEC is a supplement to the CPS that is administered between February and April each year and serves as the source of the Census Bureau's official poverty statistics and of widely used earnings measures. The sample of the CPS ASEC is the civilian noninstitutional population of the United States. There are two oversamples. The first is a sample of about 6,000 Hispanic households per year identified the previous November. The second is known as the CHIP sample, includes about 19,000 households, and was implemented to improve state estimates of Children's Health Insurance Program. I include these oversamples when I move to using the final weights in Stage 5. The oversamples bring the

[2] A more in-depth overview of the literature using administrative records to assess survey error can be found in a book chapter by Meyer and Mittag (Meyer & Mittag, 2019).

total size of the CPS ASEC to about 95,000 households per year. The CPS ASEC data include the MAFID (an ID for each address) of all addresses selected for the CPS, type of nonresponse, whether they were selected for the March supplement, whether responses to the CPS ASEC were wholly imputed due to supplement nonresponse, whether their earnings for their longest held job or other jobs were imputed, household earnings, base and final survey weights, as well as replicate weights and replicate factors.

The second source of data is the universe of 1040s filed in the United States for the 2016 to 2018 tax years other than extremely late filings. This data includes the wage and salary earnings and the address identifier or MAFID that the taxes were filed from. The CPS ASEC asks about earnings in the previous calendar year, so the 2017 to 2019 CPS ASECs are matched to the 2016 to 2018 tax years.

I collapse the CPS ASEC data and the tax data to the address level. These datasets are then merged based on the MAFID. I merge based on address because we do not have PIKs for addresses that are chosen for the CPS ASEC but do not complete the survey. Since I use MAFID to perform the merge, I restrict all of the tax data to filings that have a MAFID and a state. About 65% of addresses selected for the CPS ASEC match to the tax data. This is lower than most other studies that link tax data and the CPS ASEC, but it includes those addresses that are businesses, vacant, demolished, or don't appear in the tax data for other reasons. I count households as imputed if anyone from the household had imputed responses. The CPS ASEC has separate questions for longest held job and other jobs. I count someone as having imputed responses if either answer is imputed.

2. Method

In this paper I first calculate the earnings statistics in the tax data which has been collapsed to the address level. This gives me a benchmark to compare the same statistics calculated in CPS ASEC data. I then match the filers in the tax filings to those selected for the CPS ASEC. At each stage the households in the sample change as detailed in section 3.1. I recalculate the earnings statistics at each stage for the households that remain in the sample. For all but the eighth stage, I use the earnings from the tax data to calculate the statistics.

2.1. Stages of the Survey Production Process

As mentioned in the introduction, I compare the proxies for the survey frame to a total of eight stages in the survey production process. In the first six stages I look for the presence of generalized coverage error including unit nonresponse, then in one stage each I look at item nonresponse and measurement error. Table 1 reviews each stage of the survey production process that I explore, the type of error examined with each stage, and the weight used. More details about each stage are given below. For the first four stages I use the base weight from the survey that are meant to reflect the probability of being selected for the CPS.

- Stage 0: In order to test for bias in the earnings measures produced at each stage of the CPS sampling process, I must first find values based on the tax data to compare them to. I restrict the tax data to filings with a state and MAFID because I am using MAFID to perform the merge to the CPS ASEC.
- Stage 1: I look at tax filers who filed from an address that was selected for the CPS ASEC.
- Stage 2: I exclude type C ineligible units. This includes homes that were not interviewed because they were recorded as demolished, moved, outside segment, converted to permanent business

or storage, merged, condemned, built after April 1, 1980, on an unused line of the listing sheet, and other reasons.

- Stage 3: I exclude type B nonresponse in addition to type C. This includes homes that were not interviewed because they were vacant, temporarily occupied by persons with when their usual residence is elsewhere, unit or to be demolished, under construction and not ready, converted to a temporary business or storage, occupied by persons under fifteen, unoccupied tent or trailer site, had been granted a permit but construction had not started, and other reasons.
- Stage 4: I exclude types A, B, and C nonresponse. This includes households that refused to respond to the survey and leaves those that did respond to the CPS.
- Stage 5: I add in the oversamples for children and Hispanics at this stage. This stage corresponds to the CPS ASEC dataset if you use all the imputed responses. I also use the final March supplement weights. I continue to use these weights for the rest of the stages.
- Stage 6: I exclude households with individuals who have their entire ASEC supplement imputed (also called whole impute).
- Stage 7: I look at item nonresponse by excluding those who did not answer the earnings questions on the CPS ASEC.
- Stage 8: I switch to using the wage and salary earnings reported on the CPS ASEC and exclude households that did not report any wage and salary earnings.

2.2. Calculating Earnings Measures and Standard Errors

The earnings measures that I look at are the share of weight per decile and top five and one percent, the earnings share per decile and top one and five percent, mean earnings, and the Gini coefficient. I use calculations for earnings measures that differ from official Census reports due to pooling three years of data and streamlining calculations for computation purposes. This does not result in major differences from official earnings statistics. Following the method used in the Census's income report for the Gini coefficient, I drop all earnings values that are less than or equal to zero (Guzman & Kollar, 2023). In order to keep the samples the same across measures, I also restrict calculations of mean earnings to those with earnings that are strictly positive.

For stages five through eight I use replicate weights to calculate standard errors and test whether they are significantly different from the estimates from the full tax data.[3] For the first four stages I use replicate factors multiplied by the base weights. This is done to approximate replicate weights corresponding to base survey weights as final replicate weights are only available for respondents. More about replicate weights and factors and how they are calculated can be found in the paper by Fay and Train that first documented the Census's use of the method (Fay & Train, 1995).

I use WSI from the 1040s as my measure of earnings from the tax data. It is important to note that moving to the CPS ASEC values for the wage and salary earnings in stage eight also means dropping the values for which there is not positive wage and salary earnings reported in the CPS ASEC. I do show versions of the shares of total weight in the deciles and top one and five percent both including and excluding households that report no wage and salary earnings despite someone filing from their address

[3] Replicate weights in the CPS are a set of one hundred and sixty weights for each observation to properly account for the complex survey design.

with positive earnings. This means that my calculation of measurement error does include changes caused by both respondents saying their wage and salary earnings is negative, zero, or missing when that does not match the tax data and respondents saying their wage and salary earnings is different from what is reported in the tax data.

The CPS ASEC also includes imputed values for observations that do not respond to the CPS ASEC (dropped in stage 6) and those that do not respond to the earnings question (dropped in stage 7). I test the consequences of excluding or including those imputations when calculating earnings inequality by finding the Gini coefficient and mean earnings when using the wage and salary data from the CPS ASEC at stages 5, 6, and 7. Stages 5 and 6 include imputed values, while stage seven will be the same as stage 8 for the main WSI results. This will show whether including respondents who were either fully imputed or whose responses to the earnings question were imputed is important to reproducing the survey frame.

3. Results

I start the results by showing the impact on the shares of the weights and the earnings shares at each stage. These results help frame what I see when I look at the Gini coefficient and mean earnings by stage. Lastly, I turn to looking at the mean earnings and Gini coefficient in the CPS ASEC data including or excluding imputed values.

Table 2 shows the cumulative weight of the households chosen for the CPS ASEC at each stage of the sampling process for households within each decile and the top five and one percent. Looking at these shares allows us to examine how well the CPS ASEC is capturing tax filers over the income distribution. The shares of the weight at each stage are calculated based on the total weight of the addresses that are in the sample at that stage. The cutoffs for the deciles and top five and one percent are defined by where they fall in the tax data and kept constant when looking at the other stages. I exclude households with zero wages and salary earnings in the tax data. Here, I include two versions of the eighth stage where I switch to the responses from the CPS ASEC. First, I include those with zero earnings in the CPS ASEC data. In the second version, I exclude those that report zero earnings in the CPS ASEC.

Figure 1 shows these estimates only through stage seven or before I switch to actual CPS ASEC responses. By stage seven, when I exclude imputed responses for item nonresponse, all but three deciles are significantly different from ten percent, though my estimates for each decile are within 0.53 percentage points. We see that there is some variation in how well different deciles are picked up in the CPS ASEC. This broadens slightly through the stages. The second, third and fourth deciles have the least weight and the seventh, eighth, ninth, and tenth have the most. This shows that through the first seven stages the CPS ASEC is over weighting the top of the distribution and underweighting the lower half.

In Figures 2 and 3, I show the same graph but now include stage eight when I use the CPS ASEC responses including or excluding those households that report zero earnings, respectively. At stage eight, the share of the weight reporting earnings below the first decile is 25% when including those with no earnings (Figure 2). That dramatic increase entirely disappears when we exclude those that report not receiving any earnings (Figure 3). There are 13,000 addresses where people report zero wage and salary earnings in the CPS ASEC despite tax filings from their address having positive wage and salary

earnings.[4] This is about 20% of those households that remain at stage seven. In both figures we see that the graph widens at stage eight, indicating that CPS ASEC responses often do not align with tax filing amounts.

In Table 3, I look at where these addresses with positive WSI on tax returns and none in the CPS ASEC fall in the earnings distribution in the tax data. I find that they tend to come from the low end of the distribution. Their average earnings are $16,000 less than my total population, and over 18% of them are in the bottom decile. This issue likely drives much of the results in the rest of this analysis.

In Table 4 and Figure 4, I show the earnings shares claimed by people who report earnings within each decile and top five and one percent of the earnings distribution in the tax filings. Again, I do this by stage in the survey process. We see that the tenth decile starts with 2.14 percentage points less earnings than we see in the tax data. In Table 4, we see that in the top five and one percent we are missing 2.31 percentage points and 2.71 percentage points respectively, raising concerns that top earnings households are not being selected for the CPS ASEC. It may also be the case that the right tail of the earnings distribution is so skewed, and there are so few people in the far-right tail that the CPS ASEC is unlikely to capture earnings in that decile based on random chance. It may be necessary for the CPS ASEC to oversample high income households in order to accurately capture the income distribution.

In the fifth stage, when moving to the final survey weights and including the oversamples, we see that the estimated difference in the earnings share becomes 12.33 percentage points for the top ten percent, 12.57 percentage points for the top five percent, and 10.9 percentage points for the top one percent. This points to the possibility that the final weights are too low for households at the top end of the earnings distribution.

Table 5 and Figure 5 show the estimates for mean earnings for each of the eight stages and the tax filings. Here, as when I look at the Gini coefficient, I include only earnings that are greater than zero. This includes when I move to the CPS ASEC responses I remove addresses that do not report positive earnings. I estimate mean earnings in the tax filings to be $86,650. By stage seven, I estimate mean earnings to be only $50 different and not significantly so. In stage eight, when I move to CPS ASEC responses, I estimate mean earnings to be $6,230 higher. In Table 6, I test whether the change from each stage to the next is significant. I find that several are significant including dropping type C and type B nonresponse, excluding income nonresponse, and moving to the CPS ASEC responses. Moving to CPS ASEC responses is a significantly larger jump than any of the other changes between stages.[5] Mean earnings rising when moving to the CPS ASEC responses is consistent with more households from the bottom half of the distribution reporting zero wage and salary earnings on the CPS ASEC while reporting positive wage and salary income on their taxes.

I get results that point to a similar story when looking at the Gini coefficient in Tables 7 and 8 and Figure 6. I estimate a Gini coefficient in the tax filings of 0.4928. At all the first seven stages, I estimate a Gini coefficient that is no more than 0.0117 different from the tax filings. When I look at earnings that are reported on the CPS ASEC the Gini coefficient falls to 0.4454. This is again the largest change between stages.[6] It is also consistent with households from mostly the low end of the earnings

[4] This can be seen in the change in observations from stage seven to eight in Tables 5 or 7.
[5] This can be seen in Appendix Table 1.
[6] This is tested in Appendix Table 2.

distribution reporting no wage and salary income on the CPS ASEC and thus dropping out of the calculation as suggested by the results in Table 3.

The only other change between stages that is significant for both measures is when we drop type B nonresponse. This includes homes that were vacant, temporarily occupied by persons with their usual residence elsewhere, to be demolished, under construction and not ready, converted to a temporary business or storage, occupied by armed forces members or persons under fifteen, unoccupied tent or trailer site, or had been granted a permit but construction had not started. While this indicates that households being dropped from the survey process and marked as type B nonresponse is an issue, it is still much smaller than the measurement error seen when moving to the CPS ASEC responses. For mean earnings, we see a change of $570 at this step and a change of $6,180 when looking at measurement error (Table 6). For the Gini coefficient there is a change of 0.0039 when excluding type B nonresponse and a change of 0.0357 when moving to the CPS ASEC responses (Table 8).

Finally, I evaluate the imputations used in the CPS ASEC. In Tables 9 and 10, I show the mean earnings and Gini coefficient using wage and salary earnings from the CPS ASEC when including all imputations, excluding addresses where their responses for the CPS ASEC were fully imputed (supplement nonresponse), and excluding all imputations including those that did not respond to the earnings question (item nonresponse). In these tables, I only keep the addresses that linked to the tax data. This allows a comparison to the estimates in Tables 5 and 7 to test whether the imputations help mitigate the measurement error. I do not find a consistent story. For mean earnings the imputations don't seem to help, while for the Gini coefficient they may.

In Tables 11 and 12, I repeat Tables 9 and 10 for the full CPS ASEC sample rather than just those linked to the tax data. This is useful for public data users who do not have access to the tax data and want to know whether it's better off including the imputations or not. Large differences could also point to bias in my results coming from the effects of non-filers, tax filers who filed from a different address than where they were selected for the CPS, and filers who were mismatched. I find that Gini and mean earnings are slightly closer to estimates in the tax filings, but again do not find a consistent story on whether it's helpful to leave out imputed values.

4. Limitations of This Analysis

There are several possible sources of non-sampling error in the CPS ASEC that this research design is not well suited to separate out. These include computer processing error or problems in the code used to process the data, field processing error or mistakes made by interviewers, sampling frame omission caused by inaccuracies in the Master Address File, housing unit misclassification or type A, B, and C nonresponse being misidentified, and housing unit roster error or mistakes made in the classification of individuals who are residents of addresses selected for the CPS ASEC.[7]

Because I am using MAFIDs to merge the tax data to the CPS ASEC I restrict the tax data to just addresses with a MAFID in our data. If missing a MAFID is not random then this will result in the analysis population not resembling the whole universe of tax filers. In Table 13, I look at how addresses that are

[7] The CPS Design and Methodology Technical Paper 77 provides a good overview of potential sources of error in the CPS ASEC and the steps taken to mitigate them (US Census Bureau, 2019).

missing MAFIDs compare to the rest of the distribution. I find that about 11% of filings are missing a MAFID, and filings from PO boxes or rural routes are much more likely to be missing. Filings from the top of the earnings distribution are less likely to be missing a MAFID. This indicates that my results may reflect a population that is slightly skewed towards the top of the distribution.

The interpretation of the results from this paper depends on the accuracy of the tax data or at least the earnings measures calculated from that data. What this paper explores is how the data sources differ. This is another reason why I run the analysis on wage and salary earnings. Wage and salary earnings may be more likely to be accurate in the tax data than adjusted gross income because it is reported by both the employee and the employer. From 2011 to 2013, the Internal Revenue Service estimated the net misreporting percentage for wage, salary, and tip income to be only one percent while the same percentage for all income was eighteen percent (Internal Revenue Service, Research, Applied Analytics & Statistics, 2019). Using WSI also allows me to better assess measurement error, or how accurately survey respondents report their earnings in the CPS ASEC as the CPS ASEC does not directly ask about AGI. Still, misreporting in the tax data is a potential source of bias in these results.

Another limitation of this paper is that the administrative tax records miss the non-filing population. This means that I can only examine how well the CPS ASEC represents the portion of the sample that files taxes. There are several possible reasons why I might expect differences in how well the CPS ASEC covers the portion of the population that does not file taxes. For example, people who do not file taxes may be less likely to respond to the survey due to reluctance to report information to the government, or they may be more likely to live in dwellings not that are not included in the MAF.

A related issue is that I cannot match tax filers to the CPS ASEC data if they did not file from the address they live at. I cannot rule out that these individuals are then selected for the CPS at an address that is not the one they filed from. This could bias my findings and make us more likely to conclude that the CPS ASEC sample is not representative if these individuals are nonrandom or have different average earnings than the rest of the population.

Similarly, the linkage between the tax data and the CPS ASEC is imperfect. There is the potential for both mismatches where a CPS household is matched to incorrect tax-filers and for CPS households that are not linked to tax filers when they did file. This is mitigated to some extent because I match by address rather than by individual. To understand the effect of non-filers, tax filers who filed from a different address than where they were selected for the CPS, and filers who were mismatched, I compare all CPS ASEC households, including unmatched households to the survey proxies as well in Tables 11 and 12.

5. Conclusion

In this paper I match CPS ASEC data to tax data to study how earnings measures change throughout the survey production process. This allows a test for generalized coverage error, item nonresponse error, and measurement error in the mean earnings, Gini coefficient, and earnings and weight shares by decile and top five and one percent calculated with wage and salary earnings.

Overall, this analysis finds that there is some generalized coverage error and item nonresponse bias, but those sources of error are much smaller than that caused by measurement error. This aligns with what Meyer and Mittag found when they looked at program receipt (Meyer & Mittag, 2021). A

large portion of the measurement error is caused by households, largely from the lowest two deciles of the earnings distribution, not reporting any wage and salary earnings on the CPS ASEC despite someone filing from their address claiming positive wage and salary earnings. I also find that the sample of addresses selected for the CPS ASEC appears to miss some of the top of the distribution. This is worsened when moving to the final weights which seem to put too little weight on high earning households. I find some evidence of bias coming from type B nonresponse. Reflecting the mixed results in the previous literature, I do not see consistent results when looking at whether including imputed values improves measurement of earnings statistics.

This work shows the importance of reducing measurement error in survey responses. Researchers have explored many ways of doing this including asking respondents to look at their financial records, using conversational interview formats, improving survey questions, and having interviewers that have similar sociodemographic characteristics as the respondents (Eggleston & Reeder, 2018) (West & Blom, 2017).

There are several limitations of this research that warrant further study. I am not able to assess other sources of error such as mistakes made by interviewers, inaccuracies in the MAF, and housing unit classification. Improving the rate and accuracy at which the CPS ASEC files and the tax data can be linked would also improve this research.

Additional research should look at whether the underrepresentation of high earnings households could be due to random selection of a very skewed population. It may be the case that an oversample of high-earnings households would help the CPS ASEC better reflect the whole earnings distribution. The survey weights further decreased the difference in the income share for the top of the earnings distribution, indicating that they may need to be revised to place more weight on those households. More work should also be done to further understand when current imputation methods improve accuracy. There is also room for more research using administrative records to assess how surveys perform with other metrics besides those I look at here. Other work has also found that survey error differs within different demographics which this paper does not look at (Celhay, Meyer, & Mittag, 2021). More work could also be done to look at how survey error has changed over time.

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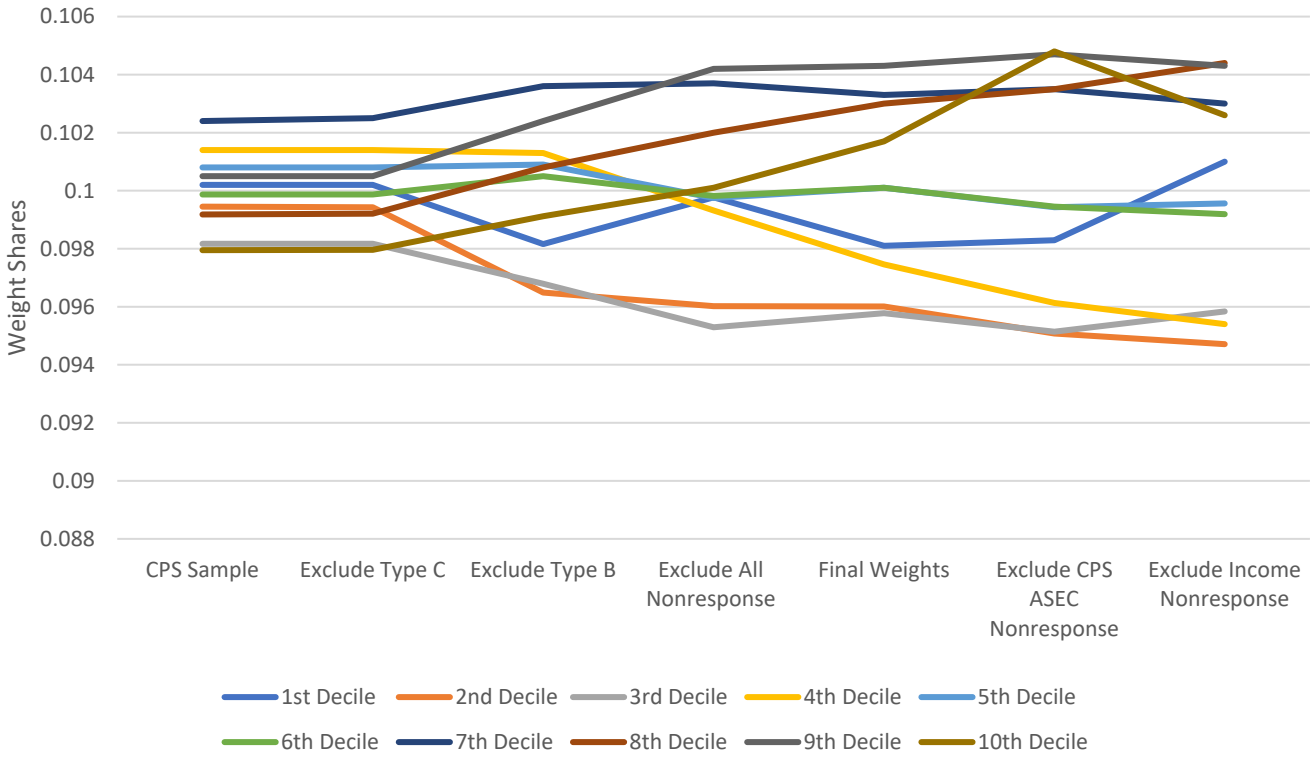
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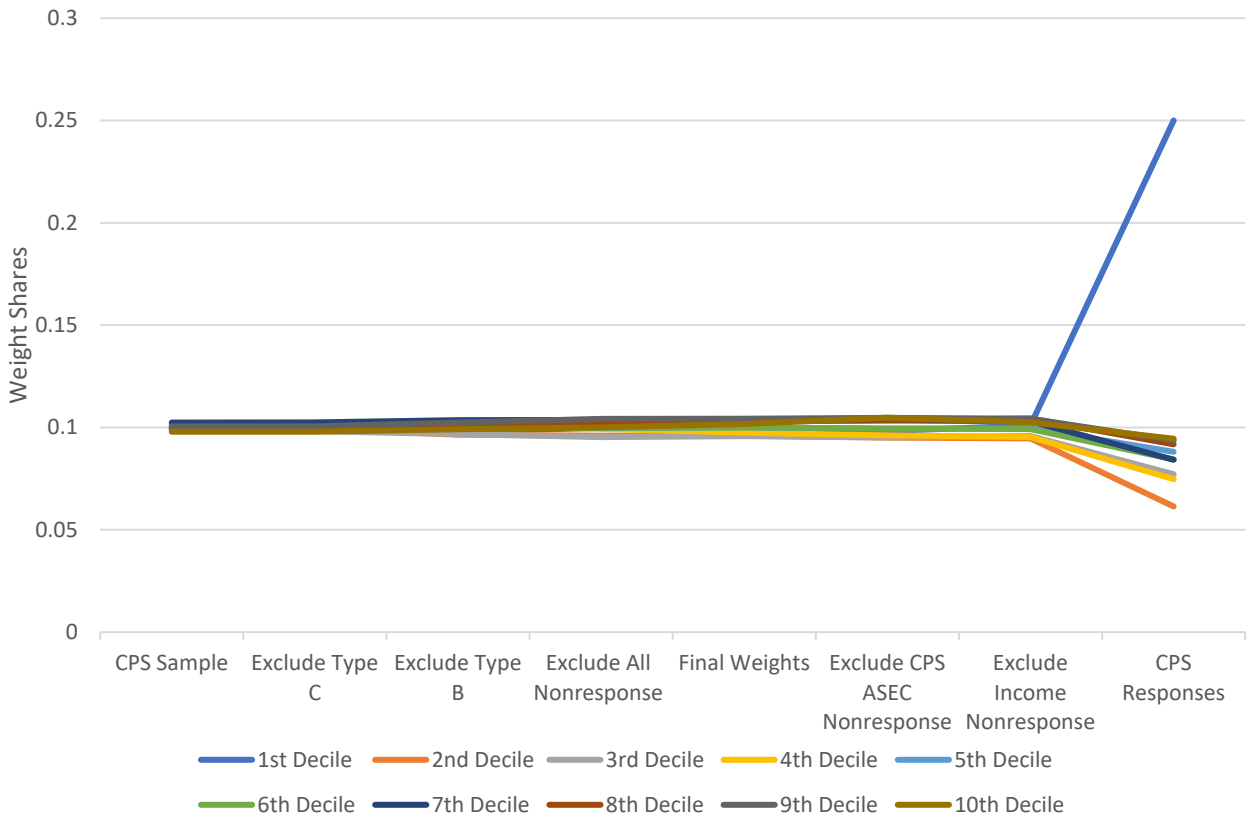
Figures and Tables

Figure 1: Weight Shares of Matched Sample for Stages 1 Through 7



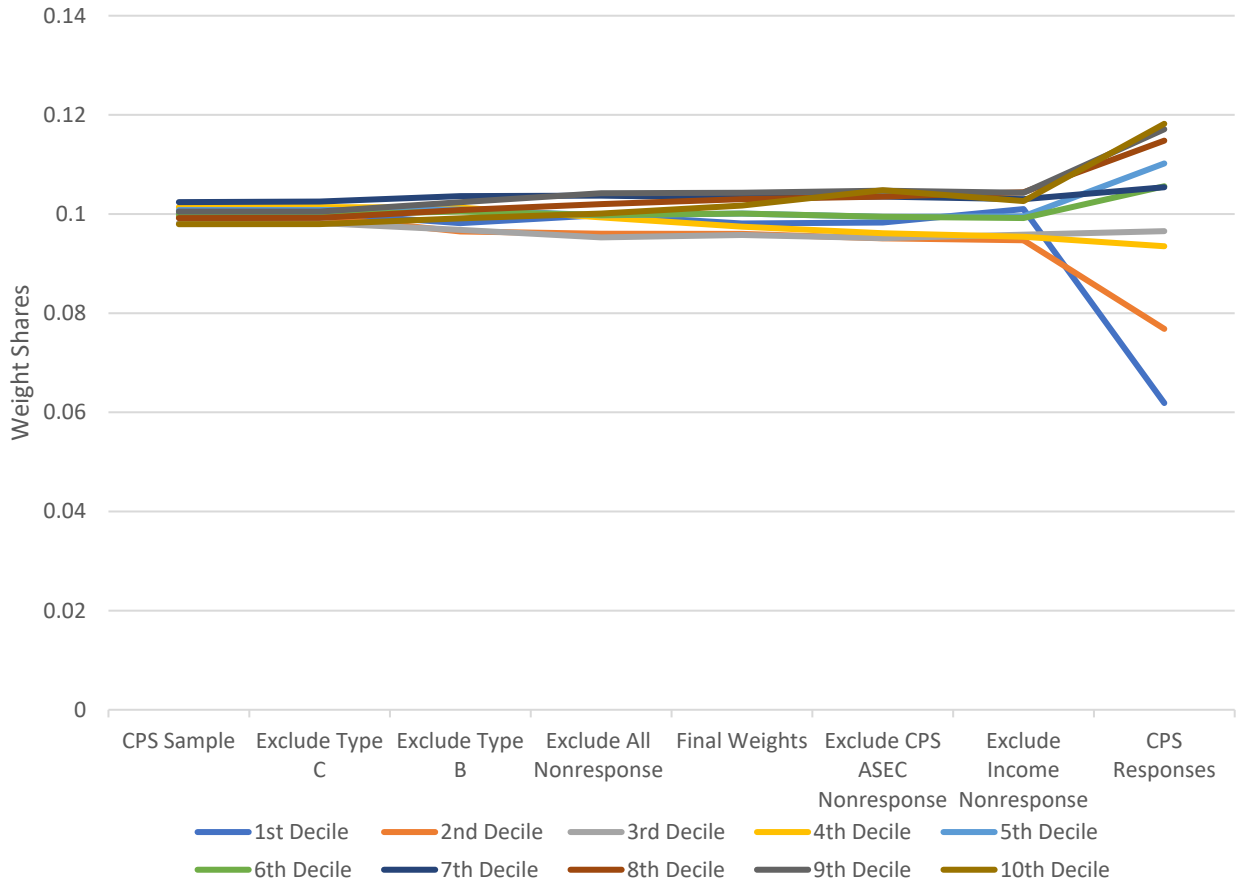
Notes: This figure shows the share of total weights of households chosen for the CPS ASEC by each decile in the tax data from seven stages in the survey production process for the CPS ASEC. The weight shares for each stage are calculated using wage and salary income from the 1040 filings except for the last stage which is calculated using the wage and salary income in the CPS ASEC. All estimates are rounded in accordance with US Census Bureau disclosure policies.
Sources: Internal Revenue Service 2016 to 2018 1040 filings matched to US Census Bureau 2017 to 2019 Current Population Survey Annual Social and Economic Supplements
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Figure 2: Weight Shares of Matched Filers by Stage Including Survey Responses for Those Reporting No Wage and Salary Income



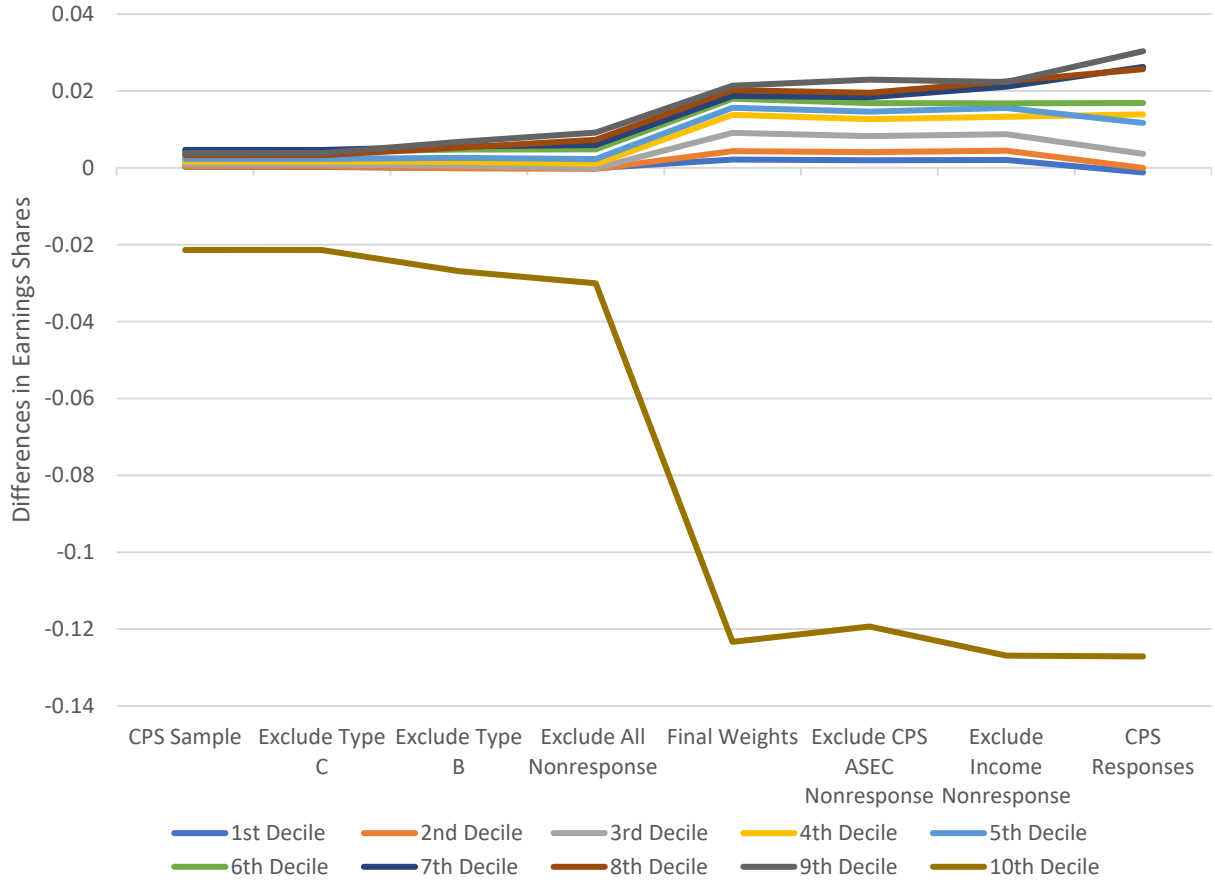
Notes: This figure shows the share of total weights of households chosen for the CPS ASEC by each decile in the tax data for eight stages in the survey production process for the CPS ASEC. The eighth stage is shown including those that report zero earnings in the CPS ASEC. The weight shares are calculated using wage and salary income from the 1040 filings except for the last one which is calculated using the wage and salary income in the CPS ASEC. All estimates are rounded in accordance with US Census Bureau disclosure policies.
Source: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplement
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Figure 3: Weight of Matched Filers by Stage Excluding Survey Responses for Those Reporting No Wage and Salary Income



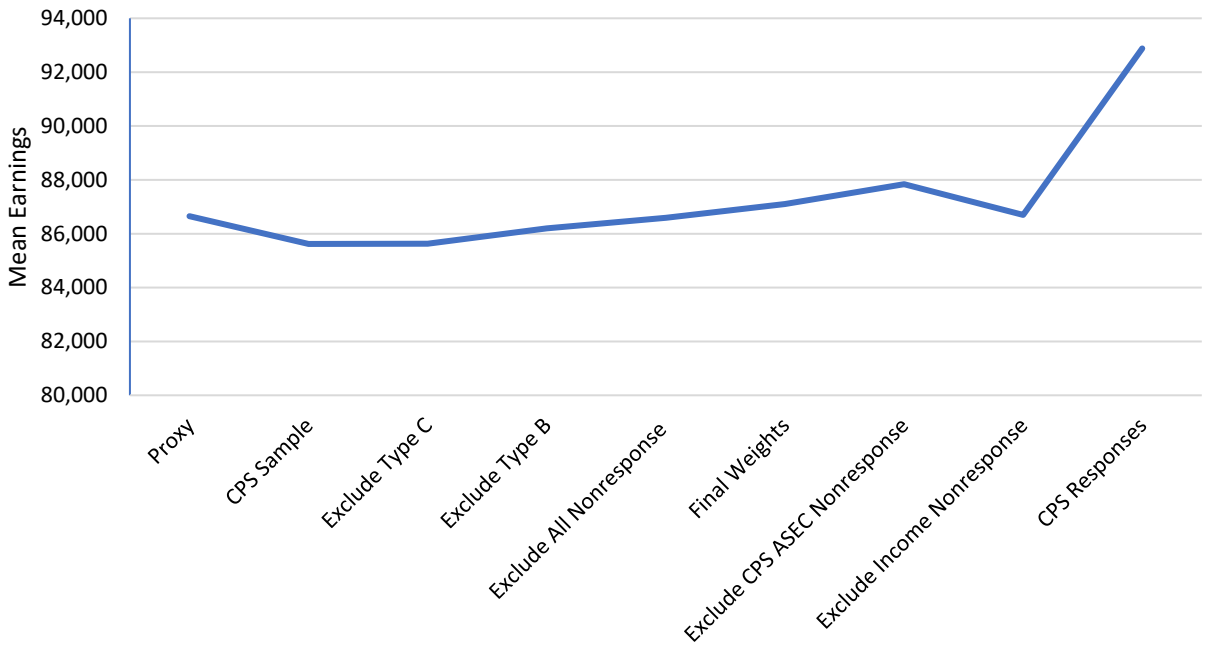
Notes: This figure shows the share of total weights of households chosen for the CPS ASEC by each decile in the tax filings for eight stages in the survey production process for the CPS ASEC. The eighth stage is shown not including those that report zero earnings in the CPS ASEC. The weight shares are calculated using wage and salary income from the 1040 filings except for the last one which is calculated using the wage and salary income in the CPS ASEC. All estimates are rounded in accordance with US Census Bureau disclosure policies.
Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplements
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Figure 4: Difference in Earnings Shares per Decile at Each Stage Compared to Tax Data



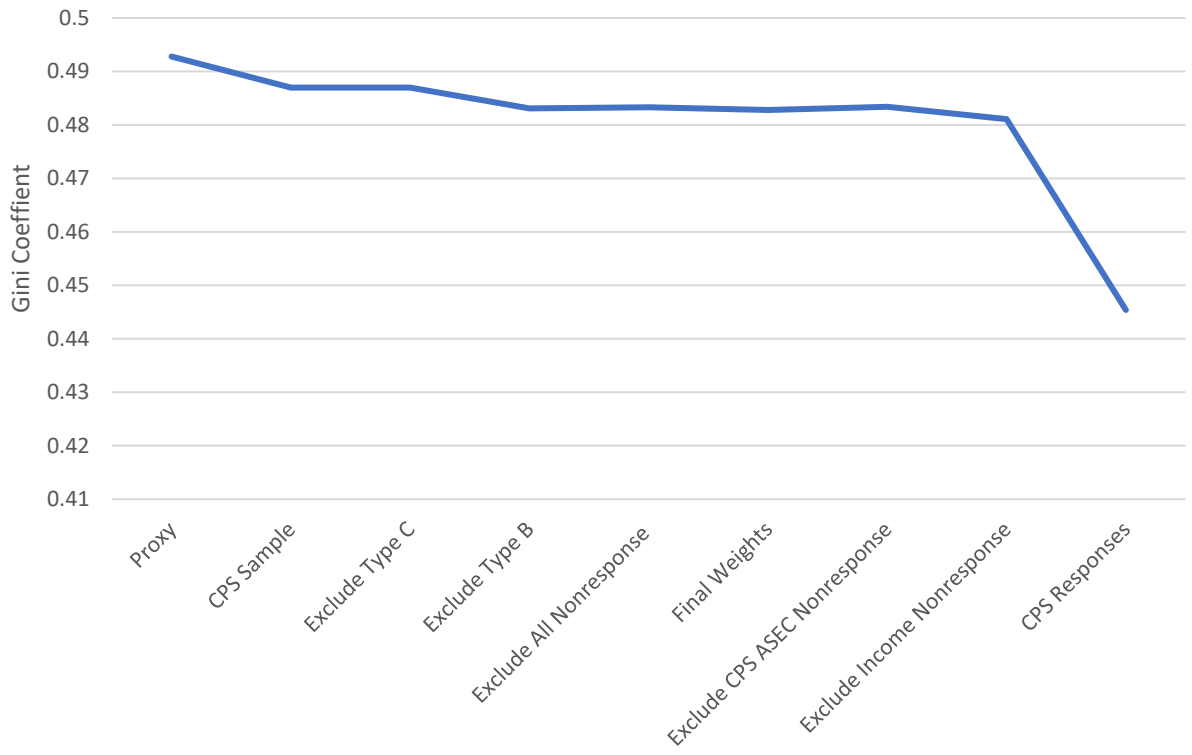
Notes: This figure shows the results comparing the earnings shares claimed by each decile in the tax filings to the earnings shares from eight stages in the survey production process for the CPS ASEC. The mean earnings are calculated using wage and salary income from the 1040 filings except for the last one which is calculated using the wage and salary income in the CPS ASEC. All estimates are rounded in accordance with US Census Bureau disclosure policies.
Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplements
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Figure 5: Mean Earnings at Each Stage



Notes: This figure shows mean earnings in the eight stages of the survey production process for the CPS ASEC. The mean earnings are calculated using wage and salary income from the 1040 filings for all stages except for the last one which uses wage and salary income in the CPS ASEC. All estimates are rounded in accordance with US Census Bureau disclosure policies.
Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplements
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Figure 6: Gini Coefficient at Each Stage



Notes: This figure shows the the Gini coefficient in the eight stages in the survey production process for the CPS ASEC. The Gini coefficients are calculated using wage and salary income from the 1040 filings except for the last one which is calculated using the wage and salary income in the CPS ASEC. All estimates are rounded in accordance with US Census Bureau disclosure policies.
Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplements
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Table 1: Stages of the Sampling Process

Stage Number	Description	Type of Error	Weight Used
0	All tax returns with a MAFID and state to serve as a comparison		None
1	Tax returns linked to the CPS ASEC	Generalized Coverage Error	Base
2	Exclude type C Nonresponse (Ineligible)	Generalized Coverage Error	Base
3	Also Exclude type B Nonresponse (Unoccupied)	Generalized Coverage Error	Base
4	Also Exclude type C Nonresponse (Refusal)	Generalized Coverage Error (Unit Nonresponse)	Base
5	Include oversamples and use final survey weights	Generalized Coverage Error	Final
6	Exclude fully imputed supplement observations	Supplement Nonresponse	Final
7	Exclude those who did not answer the income questions	Item Nonresponse	Final
8	Use earnings values from the CPS ASEC	Measurement Error	Final

Notes: This table describes the tax filing population I use as well as the eight stages in the survey production process that I compare to.

Table 2: Weight Portions by Decile and Top Percentages at Each Stage

Stage	1st Decile	2nd Decile	3rd Decile	4th Decile	5th Decile	6th Decile	7th Decile	8th Decile	9th Decile	10th Decile	Top 5%	Top 1%
CPS Sample	0.1002	0.09945	0.09817	0.1014	0.1008	0.09987	0.1024	0.09918	0.1005	0.09795	0.04942	0.01001
Difference	-0.0002	0.00055**	0.00183***	-0.0014***	-0.0008**	0.00013	-0.0024***	0.00082***	-0.0005	0.00205***	0.00058***	-0.00001
Std. Error	0.0002128	0.0002768	0.0000156	0.0003534	0.000367	0.0001408	0.0002676	0.0000578	0.0005954	0.0001349	0.0000981	6.58E-05
Exclude Type C	0.1002	0.09943	0.09817	0.1014	0.1008	0.09987	0.1025	0.09921	0.1005	0.09796	0.04944	0.01001
Difference	-0.0002	0.00057	0.00183*	-0.0014	-0.0008	0.00013	-0.0025**	0.00079	-0.0005	0.00204	0.00056	-0.00001
Std. Error	0.001161	0.001233	0.001109	0.001159	0.001161	0.001165	0.001058	0.001272	0.001225	0.001563	0.001098	0.000447
Exclude Type B	0.09816	0.09649	0.09679	0.1013	0.1009	0.1005	0.1036	0.1008	0.1024	0.09912	0.04943	0.009751
Difference	0.00184	0.00351***	0.00321***	-0.0013	-0.0009	-0.0005	-0.0036***	-0.0008	-0.0024*	0.00088	0.00057	0.000249
Std. Error	0.001179	0.001224	0.001177	0.001187	0.001197	0.001192	0.001097	0.001313	0.001274	0.001599	0.001113	0.000451
Exclude All Nonresponse	0.09977	0.09602	0.09529	0.09932	0.09975	0.09982	0.1037	0.102	0.1042	0.1001	0.04992	0.009775
Difference	0.00023	0.00398***	0.00471***	0.00068	0.00025	0.00018	-0.0037***	-0.002	-0.0042***	-0.0001	0.00008	0.000225
Std. Error	0.001289	0.001331	0.001268	0.001299	0.001267	0.001251	0.001132	0.001402	0.001394	0.001673	0.001171	0.000499
Final Weights	0.0981	0.09601	0.09578	0.09746	0.1001	0.1001	0.1033	0.103	0.1043	0.1017	0.05049	0.009779
Difference	0.0019	0.00399	0.00422*	0.00254	-0.0001	-0.0001	-0.0033*	-0.003*	-0.0043	-0.0017	-0.00049	0.000221
Std. Error	0.002839	0.002603	0.002555	0.002546	0.00151	0.001091	0.001739	0.001628	0.002968	0.005424	0.003255	0.000868
Exclude CPS ASEC Nonresponse	0.09829	0.09508	0.09514	0.09613	0.09943	0.09945	0.1035	0.1035	0.1047	0.1048	0.05193	0.01023
Difference	0.00171	0.00492***	0.00486***	0.00387***	0.00057	0.00055	-0.0035***	-0.0035**	-0.0047***	-0.0048***	-0.00193*	-0.00023
Std. Error	0.001203	0.001179	0.001247	0.001256	0.001341	0.001193	0.001193	0.001368	0.001282	0.001436	0.001045	0.000439
Exclude Income Nonresponse	0.101	0.09471	0.09584	0.0954	0.09956	0.09919	0.103	0.1044	0.1043	0.1026	0.05044	0.009648
Difference	-0.001	0.00529***	0.00416***	0.0046***	0.00044	0.00081	-0.003**	-0.0044***	-0.0043***	-0.0026*	-0.00044	0.000352
Std. Error	0.00149	0.001479	0.001567	0.001371	0.001536	0.001353	0.001351	0.001523	0.00143	0.001535	0.001135	0.000448
CPS Responses	0.25	0.06142	0.07717	0.07475	0.08811	0.08441	0.08423	0.09178	0.09361	0.09451	0.04431	0.007268
Difference	-0.15***	0.03858***	0.02283***	0.02525***	0.01189***	0.01559***	0.01577***	0.00822***	0.00639***	0.00549***	0.00569***	0.002732***
Std. Error	0.002404	0.001104	0.001332	0.001149	0.001276	0.001434	0.001247	0.001305	0.001372	0.001412	0.001073	0.000386
CPS Responses No Zeros	0.06187	0.07683	0.09653	0.0935	0.1102	0.1056	0.1054	0.1148	0.1171	0.1182	0.05543	0.009091
Difference	0.03813***	0.02317***	0.00347**	0.0065***	-0.0102***	-0.0056***	-0.0054***	-0.0148***	-0.0171***	-0.0182***	-0.00543***	0.000909*
Std. Error	0.00127	0.001351	0.001629	0.001382	0.001558	0.001776	0.001581	0.001596	0.001656	0.001761	0.001331	0.000478

Notes: This table shows the share of total weights of households chosen for the CPS ASEC by each decile and the top one and five percent in the tax filings from eight stages in the survey production process for the CPS ASEC. The eighth stage is shown both including those that report zero earnings in the CPS ASEC and not including those that report zero income in the CPS ASEC. The weight shares are calculated using wage and salary income from the 1040 filings except for the last one which is calculated using the wage and salary income in the CPS ASEC. The differences rows give the percent it should represent minus the shares for each stage. All estimates are rounded in accordance with US Census Bureau disclosure policies.

** p<0.10, ** p<0.05, *** p<0.01*

Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplement

The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Table 3: Distribution of Addresses with Zero CPS ASEC Wages and Positive IRS Wages

	Mean	1st Decile	2nd Decile	3rd Decile	4th Decile	5th Decile	6th Decile	7th Decile	8th Decile	9th Decile	10th Decile	Top 5%	Top 1%
Estimate	70,650	0.1837	0.1225	0.1048	0.09985	0.0959	0.08791	0.08637	0.08164	0.06935	0.06802	0.03143	0.006502
Difference	16000***	0.0837***	0.0225***	0.0048	-0.00015	-0.0041	-0.01209***	-0.01363***	-0.01836***	-0.03065***	-0.03198***	-0.01857***	-0.003498***
Standard Err.	1,807	0.004049	0.003229	0.003324	0.003047	0.00274	0.002985	0.002804	0.003083	0.002664	0.002609	0.001909	0.000812

Notes: This table shows the mean earnings and share of total weights of households with positive wage and salary income in tax filings and no wage and salary income in the CPS ASEC by each decile and the top one and five percent. The difference row gives the difference between the tax data and the households with no wage and salary income in the CPS ASEC. All estimates are rounded in accordance with US Census Bureau disclosure policies.

** p<0.10, ** p<0.05, *** p<0.01*

Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplement

The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Table 4: Earnings Shares by Decile and Top Percentages at Each Stage

	1st Decile	2nd Decile	3rd Decile	4th Decile	5th Decile	6th Decile	7th Decile	8th Decile	9th Decile	10th Decile	Top 5%	Top 1%
Tax Data	0.008256	0.02	0.03116	0.04303	0.05613	0.07096	0.0885	0.1115	0.1495	0.421	0.3159	0.1745
CPS Sample	0.008504	0.02035	0.03203	0.04466	0.05848	0.07507	0.09315	0.1148	0.1533	0.3996	0.2928	0.1474
Difference	-0.000248	-0.00035	-0.00087	-0.00163**	-0.00235**	-0.00411***	-0.00465***	-0.0033*	-0.0038*	0.0214***	0.0231***	0.0271***
Std. Error	0.0001714	0.0003986	0.000608	0.00077	0.000972	0.001326	0.001435	0.001694	0.002301	0.007524	0.008598	0.009148
Exclude Type C	0.008495	0.02035	0.03202	0.04466	0.05848	0.07508	0.09315	0.1148	0.1533	0.3996	0.2928	0.1474
Difference	-0.000239	-0.00035	-0.00086	-0.00163**	-0.00235**	-0.00412***	-0.00465***	-0.0033*	-0.0038*	0.0214***	0.0231***	0.0271***
Std. Error	0.0001714	0.000399	0.000608	0.000771	0.000972	0.001323	0.001433	0.001696	0.002302	0.007522	0.008596	0.009145
Exclude Type B	0.008203	0.01988	0.03171	0.04458	0.05873	0.07572	0.09407	0.1167	0.1562	0.3941	0.2859	0.1399
Difference	0.000053	0.00012	-0.00055	-0.00155**	-0.0026***	-0.00476***	-0.00557***	-0.0052***	-0.0067***	0.0269***	0.03***	0.0346***
Std. Error	0.0001685	0.0003837	0.000598	0.000768	0.000979	0.001348	0.001443	0.001735	0.002362	0.007491	0.008571	0.009115
Exclude All Nonresponse	0.008174	0.01973	0.03104	0.04404	0.05843	0.07576	0.09443	0.1188	0.1586	0.3909	0.2809	0.1339
Difference	0.000082	0.00027	0.00012	-0.00101	-0.0023**	-0.0048***	-0.00593***	-0.0073***	-0.0091***	0.0301***	0.035***	0.0406***
Std. Error	0.0001672	0.0003667	0.000591	0.000786	0.000986	0.001383	0.001443	0.001811	0.002465	0.007314	0.008434	0.008887
Final Weights	0.01041	0.02436	0.04026	0.05683	0.07176	0.08894	0.1072	0.1317	0.1709	0.2977	0.1902	0.06551
Difference	-0.002154***	-0.00436***	-0.0091***	-0.0138***	-0.01563***	-0.01798***	-0.0187***	-0.0202***	-0.0214***	0.1233***	0.1257***	0.109***
Std. Error	0.0005803	0.001172	0.002251	0.002598	0.002366	0.00195	0.002096	0.002585	0.003288	0.01242	0.01064	0.008463
Exclude CPS ASEC Nonresponse	0.01021	0.0241	0.03942	0.0557	0.07075	0.0878	0.1069	0.131	0.1724	0.3017	0.1926	0.06628
Difference	-0.001954***	-0.0041***	-0.00826***	-0.01267***	-0.01462***	-0.01684***	-0.0184***	-0.0195***	-0.0229***	0.1193***	0.1233***	0.1082***
Std. Error	0.0001736	0.0003666	0.000588	0.000856	0.000999	0.001157	0.001523	0.001901	0.002311	0.005388	0.005696	0.005256
Exclude Income Nonresponse	0.01032	0.02446	0.03992	0.0563	0.07173	0.08774	0.1096	0.134	0.1718	0.2941	0.1853	0.06008
Difference	-0.002064***	-0.00446***	-0.00876***	-0.01327***	-0.0156***	-0.01678***	-0.0211***	-0.0225***	-0.0223***	0.1269***	0.1306***	0.1144***
Std. Error	0.0001953	0.0004355	0.000701	0.000937	0.001175	0.001355	0.001793	0.002034	0.002447	0.005284	0.005483	0.005069
CPS Responses	0.007053	0.01999	0.0348	0.05694	0.06781	0.08785	0.1147	0.1371	0.1798	0.2938	0.1712	0.05113
Difference	0.001203***	0.00001	-0.00364***	-0.01391***	-0.01168***	-0.01689***	-0.0262***	-0.0256***	-0.0303***	0.1272***	0.1447***	0.1234***
Std. Error	0.0001501	0.0003922	0.000672	0.000924	0.001229	0.001428	0.001909	0.001959	0.002724	0.005055	0.004896	0.004033

Notes: This table shows the results comparing the earnings shares claimed by each decile and the top one and five percent in the tax filings to the earnings shares from eight stages in the survey production process for the CPS ASEC. The mean earnings are calculated using wage and salary income from the 1040 filings except for the last one which is calculated using the wage and salary income in the CPS ASEC. The mean differences rows give the value of the earnings shares from the tax filings minus the earnings shares for each stage. All estimates are rounded in accordance with US Census Bureau disclosure policies.

** p<0.10, ** p<0.05, *** p<0.01*

Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplement

The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Table 5: Mean Earnings at Each Stage

Stage	Mean	Mean Difference	Standard Error	Observations
Proxy	86,650			2.35E+08
CPS Sample	85,620	1030	687.7	108,000
Exclude Type C	85,630	1020	687.6	108,000
Exclude Type B	86,200	450	702.6	102,000
Exclude All Nonresponse	86,590	60	751.8	85,000
Final Weights	87,100	-450	2385	126,000
Exclude CPS ASEC Nonresponse	87,840	-1190*	623.2	91,000
Exclude Income Nonresponse	86,700	-50	605.8	66,000
Using CPS ASEC Income	92,880	-6230***	587.2	53,000

Notes: This table shows mean earnings in the tax filings to the mean earnings from eight stages in the survey production process for the CPS ASEC. The mean earnings are calculated using wage and salary income from the 1040 filings except for the last one which is calculated using the wage and salary income in the CPS ASEC. The Mean Difference column gives the value of the mean earnings from the tax filings (Proxy) minus the mean earnings for each stage. All estimates are rounded in accordance with US Census Bureau disclosure policies.
** p<0.10, ** p<0.05, *** p<0.01*
Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplement
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Table 6: Comparison to Previous Stage, Mean Earnings

Stages Compared	Stage Difference	Standard Error
1 minus 0	1030	687.7
2 minus 1	10**	8.147
3 minus 2	570***	104.6
4 minus 3	390*	233.8
5 minus 4	510	2323
6 minus 5	740	765.4
7 minus 6	1140***	425.7
8 minus 7	6180***	600.6

Notes: This table shows the difference between mean earnings at each stage of survey production process and the prior stage. A description of the stages can be seen in Table 1. Mean earnings are calculated using wage and salary income from the 1040 filings except for the last one which is calculated using the wage and salary income in the CPS ASEC. All estimates are rounded in accordance with US Census Bureau disclosure policies.
** p<0.10, ** p<0.05, *** p<0.01*
Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplement
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Table 7: Comparing Each Stage to the Tax filings, Gini Coefficient

Stage	Gini	Gini Difference	Standard Error	Observations
Proxy	0.4928			2.35E+08
CPS Sample	0.4870	0.0058*	0.003163	108,000
Exclude Type C	0.4870	0.0058*	0.003169	108,000
Exclude Type B	0.4831	0.0097***	0.003265	102,000
Exclude All Nonresponse	0.4833	0.0095***	0.003536	85,000
Final Weights	0.4828	0.01***	0.003768	126,000
Exclude CPS ASEC Nonresponse	0.4834	0.0094***	0.003053	91,000
Exclude Income Nonresponse	0.4811	0.0117***	0.00292	66,000
Using CPS ASEC Income	0.4454	0.0474***	0.002469	53,000

Notes: This table shows the Gini coefficient in the eight stages in the survey production process for the CPS ASEC. The Gini coefficients are calculated using wage and salary income from the 1040 filings except for the last one which is calculated using the wage and salary income in the CPS ASEC. The first line gives the Gini coefficient for the tax filings. The Gini Difference column gives the value of the Gini coefficient from the tax filings (Proxy) minus the Gini coefficient for each stage. All estimates are rounded in accordance with US Census Bureau disclosure policies.
** p<0.10, ** p<0.05, *** p<0.01*
Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplement
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Table 8: Comparison to Previous Stage, Gini Coefficient

Stages Compared	Stage Difference	Standard Error
1 minus 0	-0.0058*	0.003163
2 minus 1	0	2.84E-05
3 minus 2	-0.0039***	0.00048
4 minus 3	0.0002	0.000968
5 minus 4	-0.0005	0.001131
6 minus 5	0.0006	0.001576
7 minus 6	-0.0023	0.002225
8 minus 7	-0.0357***	0.003432

Notes: This table shows the difference between the Gini coefficient at each stage and the prior stage. A description of the stages can be seen in Table 1. The Gini coefficient is calculated using wage and salary income from the 1040 filings except for the last one which is calculated using the wage and salary income in the CPS ASEC. All estimates are rounded in accordance with US Census Bureau disclosure policies.
** p<0.10, ** p<0.05, *** p<0.01*
Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplement
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Table 9: CPS Response Comparisons for Those Linked to the Tax Data, Mean Earnings

Stage	Mean Earnings	Mean Difference	Standard Error	Observations
Proxy	86,650			2.35E+08
Final Weights	94,020	-7370***	464.4	97,500
Exclude CPS ASEC Nonresponse	93,280	-6630***	502.3	77,500
Exclude Income Nonresponse	92,880	-6230***	587.2	53,000

Notes: This table shows the results comparing the mean earnings in the tax filings to the mean earnings calculated with the responses to the CPS ASEC. The mean earnings for the proxy is calculated using wage and salary income from the 1040 filings. The mean earnings for the stages are calculated using household wage and salary earnings from the CPS ASEC. The sample for all stages is restricted to households that were matched to the tax filings and who were not removed due to nonresponse. The first line (Proxy) gives the mean earnings for the tax filings. The Mean Difference

column gives the value of the mean earnings from the tax filings (Proxy) minus the mean earnings for each stage. All estimates are rounded in accordance with US Census Bureau disclosure policies.
** p<0.10, ** p<0.05, *** p<0.01*
Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplement
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)
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Table 10: CPS Response Comparisons for Those Linked to the Tax Data, Gini Coefficient

Stage	Gini	Gini Difference	Standard Error	Observations
Proxy	0.4928			2.35E+08
Final Weights	0.4479	0.0449***	0.001693	97,500
Exclude CPS ASEC Nonresponse	0.4477	0.0451***	0.001968	77,500
Exclude Income Nonresponse	0.4454	0.0474***	0.002469	53,000

Notes: This table shows the results comparing the Gini coefficient in the tax filings to the Gini coefficients calculated with the responses to the CPS ASEC. The Gini coefficient for the proxy is calculated using wage and salary income from the 1040 filings. The Gini coefficients for the stages are calculated using household wage and salary earnings from the CPS ASEC. The sample for all stages is restricted to households that were matched to the tax filings and who were not removed due to nonresponse. The Gini Difference column gives the value of the Gini coefficient from the tax filings (Proxy) minus the Gini coefficient for each stage. All estimates are rounded in accordance with US Census Bureau disclosure policies.
** p<0.10, ** p<0.05, *** p<0.01*
Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplement
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Table 11: CPS Response Comparisons for The Full CPS ASEC Sample, Mean Earnings

Stage	Mean Earnings	Mean Difference	Standard Error	Observations
Proxy	86,650			2.35E+08
Final Weights	89,160	-2510***	385.7	140,000
Exclude CPS ASEC Nonresponse	88,380	-1730***	412.9	111,000
Exclude Income Nonresponse	87,740	-1090**	475.5	76,000

Notes: This table shows the results comparing the mean earnings in the tax filings to the mean earnings calculated with the responses to the CPS ASEC. The mean earnings for the proxy is calculated using wage and salary income from the 1040 filings. The mean earnings for the stages are calculated using household wage and salary earnings from the CPS ASEC. The sample for all stages includes households that were not matched to the tax filings. The Mean Difference column gives the value of the mean earnings from the tax filings (Proxy) minus the mean earnings for each stage. All estimates are rounded in accordance with US Census Bureau disclosure policies.
** p<0.10, ** p<0.05, *** p<0.01*
Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplement
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Table 12: CPS Responses Comparisons for the Full CPS ASEC Sample, Gini Coefficient

Stage	Gini	Gini Difference	Standard Error	Observations
Proxy	0.4928			2.35E+08
Final Weights	0.4570	0.0358***	0.001466	140,000
Exclude CPS ASEC Nonresponse	0.4570	0.0358***	0.001723	111,000
Exclude Income Nonresponse	0.4553	0.0375***	0.002154	76,000

Notes: This table shows the results comparing the Gini coefficient in the tax filings to the Gini coefficient calculated with the responses to the CPS ASEC. The Gini coefficient for the proxy is calculated using wage and salary income from the 1040 filings. The Gini coefficients for the stages are calculated using household wage and salary earnings from the CPS ASEC. The sample for all stages includes households that were not matched to

the tax filings. The Gini Difference column gives the value of the Gini coefficient from the tax filings (Proxy) minus the Gini coefficient for each stage. All estimates are rounded in accordance with US Census Bureau disclosure policies.
** p<0.10, ** p<0.05, *** p<0.01*
Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplement
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Table 13: Percentage of Tax Filings Missing MAFID

Subset	Percent Missing MAFID
All tax filings	11%
PO Boxes	61%
Rural Routes	74%
No WSI	11%
1pt WSI	11%
5pt WSI	11%
95pt WSI	6%
99pt WSI	6%

Notes: This table shows the percentage of tax filings that are missing a MAFID (address ID) by various characteristics. The characteristics include all tax filings, tax filings which list their address as a PO box, tax filings from a rural route, tax filings with no wage and salary income, and tax filings in the 1st, 5th, 95th, and 99th percentiles.
Source: Internal Revenue Service 2016 to 2018 1040 filings
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Appendix

Here I test whether the change from stage seven to stage eight, when I switch from using the tax data to estimate the statistics to using the CPS ASEC responses, is indeed the largest of the changes between the stages. I do this by using calculating the differences between the change with each set of replicate weights to test for whether the change from seven to eight is larger than the jump from each of the other stages. I find that for both the mean and the Gini coefficient the change to stage eight is statistically significantly larger than the other changes between stages.

Appendix Table 1: Comparing the Jump to Stage 8 to Other Jumps, Mean Earnings

Jumps in Stages Compared	Difference in Jumps	Standard Error
8 minus 1	5150***	761.5
8 minus 2	6170***	601.3
8 minus 3	5610***	620.1
8 minus 4	5790***	670.7
8 minus 5	5670**	2393
8 minus 6	5440***	946.1
8 minus 7	5040***	667.2

Notes: This table compares the jump in the mean earnings from stage seven to eight of the survey production process to each of the other jumps from stage zero (the tax filings) to stage seven. A breakdown of the stages can be seen in Table 1. All estimates are rounded in accordance with US Census Bureau disclosure policies.
** p<0.10, ** p<0.05, *** p<0.01*
Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplement
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)

Appendix Table 2: Comparing the Jump to Stage 8 to Other Jumps, Gini Coefficient

Jumps in Stages Compared	Difference in Jumps	Standard Error
8 minus 1	0.0299***	0.005592
8 minus 2	0.4454***	0.003433
8 minus 3	-0.0039***	0.003487
8 minus 4	-0.0002***	0.003501
8 minus 5	-0.0005***	0.003687
8 minus 6	-0.0006***	0.003805
8 minus 7	-0.0023***	0.00448

Notes: This table compares the jump in the Gini coefficient from stage seven to eight of the survey production process to each of the other jumps from stage zero (the tax filings) to stage seven. A breakdown of the stages can be seen in Table 1. All estimates are rounded in accordance with US Census Bureau disclosure policies.
** p<0.10, ** p<0.05, *** p<0.01*
Sources: Internal Revenue Service 2016 to 2018 1040 filings and the 2017 to 2019 Current Population Survey Annual Social and Economic Supplement
The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7503840 (7505713): CBDRB-FY23-CES014-029, CBDRB-FY24-CES026-012)